

Effect of ascorbic acid and green tea on endogenous formation of N-nitrosodimethylamine and N-nitrosopiperidine in humans.



Many constituents present in the human diet may inhibit endogenous formation of N-nitroso compounds (NOC). Studies with human volunteers showed inhibiting effects of intake of ascorbic acid and green tea consumption on nitrosation using the N-nitrosoproline test. The aim of the present study was to evaluate the effects of ascorbic acid and green tea on urinary excretion of carcinogenic N-nitrosodimethylamine (NDMA) and N-nitrosopiperidine (NPIP) in humans. Twenty-five healthy female volunteers consumed a fish meal rich in amines as nitrosatable precursors in combination with intake of nitrate-containing drinking water at the Acceptable Daily Intake level during 7 consecutive days. During 1 week before and after nitrate intake a diet low in nitrate was consumed. Using the same protocol, the effect of two different doses of ascorbic acid (250 mg and 1 g/day) and two different doses of green tea (2 g and 4 g/day) on formation of NDMA and NPIP was studied. Mean nitrate excretion in urine significantly increased from control (76 $\pm$ 24) to 167 $\pm$ 25 mg/24 h. Intake of nitrate and fish resulted in a significant increase in mean urinary excretion of NDMA compared with the control weeks: 871 $\pm$ 430 and 640 $\pm$ 277 ng/24 h during days 1-3 and 4-7, respectively, compared with 385 $\pm$ 196 ng/24 h (p <0.0002). Excretion of NPIP in urine was not related to nitrate intake and composition of the diet. Intake of 250 mg and 1 g of ascorbic acid per day resulted in a significant decrease in urinary NDMA excretion during days 4-7 (p =0.0001), but not during days 1-3. Also, consumption of four cups of green tea per day (2 g) significantly decreased excretion of NDMA during days 4-7 (p =0.0035), but not during days 1-3. Surprisingly, consumption of eight cups of green tea per day (4 g) significantly increased NDMA excretion during days 4-7 (p =0.0001), again not during days 1-3. This increase is probably a result of catalytic effects of tea polyphenols on nitrosation, or of another, yet unknown, mechanism. These results suggest that intake of ascorbic acid and moderate consumption of green tea can reduce endogenous NDMA formation.